

ORIGINAL ARTICLE

Effect of substituting fishmeal with corn protein concentrate on growth performance, nutrient utilization and skin coloration in red hybrid tilapia, *Oreochromis* sp.

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Abstract

The objective of this study was to evaluate the effects of fishmeal (FM) replacement with corn protein concentrate (CPC) on growth performance, nutrient utilization, gut morphology and skin coloration of red hybrid tilapia, *Oreochromis* sp. Five isonitrogenous (350 g/kg crude protein) and isolipidic (10 g/kg lipid) diets were formulated to contain CPC that substituted 0%, 25%, 50%, 75% or 100% FM. Diets were fed to triplicate groups of tilapia (mean initial weight, 10.33 ± 0.02 g) twice daily for 63 days. The results showed that replacing up to 50% FM in red hybrid tilapia diet with CPC did not show any significant adverse effects on growth, feed utilization, haematocrit counts, condition factor and gut morphology of tilapia ($p > 0.05$). However, replacing 75% or 100% FM with CPC had deleterious effects ($p < 0.05$). Carotenoids in CPC contributed to skin yellowness, which was significantly higher in the diet where 100% FM was replaced with CPC. Using regression analysis, the optimal substitution level of FM by CPC was estimated at 25% for percentage weight gain, 33% for FCR and 29% for protein efficiency ratio. CPC could be used as a single plant protein source to substitute up to 50% FM in red hybrid tilapia diets.

KEYWORDS

corn protein concentrate, fish colour, fishmeal, growth, plant protein, tilapia

1 | INTRODUCTION

Numerous studies have been conducted in recent years on the use of alternative plant ingredients as protein sources to substitute fishmeal (FM) in aquafeeds, in order to reduce the reliance on fishery-derived products and to improve cost efficiencies. Plant protein sources such as soybean meal (SBM) (El-Ebiary, 2005; Goda, Wafa, El-Haroun, & Chowdbury, 2007), cassava leaf meal (Ng & Wee, 1989), corn gluten meal (CGM) (El-Ebiary, 2005; Wu, Rosati, Sessa, & Brown, 1995), cottonseed meal (Mbahinzireki, Dabrowski, Lee, El-Saidy, & Wisner, 2001), rapeseed meal (Davies, McConnell, & Bateson, 1990) and many others, as reviewed by Ng and Romano (2013), have been evaluated for tilapia feeds and for use in other aquafeeds (Gatlin et al., 2007).

Despite their availability and lower costs, the major constraints with the use of plant-based protein sources are the imbalanced amino acid profile as well as the presence of anti-nutritional factors (ANF) in some of these ingredients. According to Sanchez-Lozano, Martinez-Llorens, Tomas-Vidal, and Cerda (2009), cereals (corn, wheat, barley and rice) are generally deficient in lysine, while legumes (rapeseed, soy, alfalfa and pea) are deficient in methionine and cysteine. The presence of ANF such as saponins, tannins, phorbol esters and gossypol in plant protein sources is likely to compromise fish growth and health (Borgeson, Racz, Wilkie, White, & Drew, 2006; Chen et al., 2011; Francis, Makkar, & Becker, 2001; Sorenson et al., 2011). To enhance nutritional quality and eliminate ANF, further processing of plant ingredients can be conducted to produce plant protein

concentrates. Soy (Zhao et al., 2010), potato (Tusche et al., 2013), rice (Oujifard, Seyfabadi, Kenari, & Rezaei, 2012), lupin and pea (Zhang et al., 2012) and rapeseed (Slawski et al., 2012) protein concentrates have been successfully used at greater inclusion levels in fish feeds. Furthermore, the supplementation of limiting amino acids is a good way to balance the essential amino acid profile of these concentrates and improve their nutritional value. Thus, a relatively new plant protein concentrate is lysine-enhanced corn protein concentrate (CPC, Lysto[®]; Cargill Corn Milling; Cargill Inc.), produced from corn through an enzymatic process that contains high crude protein and lysine levels (52 g/kg; Table 1), low ash and starch levels and no known ANF. Typical proximate composition (as is basis) per kg of CPC is 756 g protein, 47 g crude fat, 7 g crude fibre, 13 g ash and 53 g carbohydrates (Eric Bell, Cargill Corn Milling, Cargill Inc; pers. comm.). This ingredient is golden yellow in colour due to the presence of naturally occurring carotenoids in the corn product. To date, limited published information is available on the use of CPC in aquafeeds. Non-lysine-enhanced CPC (Empyreal[®]75; Cargill Corn Milling) with lysine content of about 11 g/kg (as is basis) had been evaluated as part of various protein blends for replacement of animal protein sources in the diets of Atlantic salmon *Salmo salar* (Burr, Wolters, Barrows, & Hardy, 2012) and Florida Pompano *Trachinotus carolinus* (Novriadi, Spangler, & Davis, 2019). High lysine CPC (Lysto[®]) has been reported for use as a protein blend in the diets of Florida Pompano (Cook, Zhou, Rhodes, & Davis, 2016) and as a direct replacement of dietary FM in the diets of Nile tilapia, *Oreochromis niloticus* (Khalifa, Belal, El-Tarabily, Tariq, & Kassab, 2018).

TABLE 1 Amino acid profile of corn protein concentrate

Amino acid	Protein ^a (g/kg)
Alanine	81.6
Arginine	30.2
Aspartic acid	56.2
Cystine	15.2
Glutamic acid	202.6
Glycine	23.7
Histidine	20.6
Isoleucine	40.6
Leucine	167.3
Lysine	52.0
Methionine	21.6
Phenylalanine	61.6
Proline	88.4
Serine	46.0
Threonine	29.8
Tryptophan	4.2
Tyrosine	52.4
Valine	44.5

^aTypical amino acid profile of corn protein concentrate with the exception of lysine which is enhanced in Lysto[®], as provided by the manufacturer (Cargill Corn Milling, Cargill Inc.).

The farming of red hybrid tilapia, *Oreochromis* sp., is very popular in Asia, thanks to its resemblance (coloration) to highly priced marine fish species such as sea bream *Chrysophrys major* and red snapper *Lutjanus campechanus* (Watanabe, Ernst, Olla, & Wicklund, 1989), which increases its market acceptance and value. Although it was reported that half of the dietary FM could be replaced in Nile tilapia diet with CPC (Khalifa et al., 2018), this has yet to be validated with red hybrid tilapia, especially in the context that CPC could have a negative effect on the skin colour, which play an important role in its market attractiveness. Furthermore, some aquafeed millers are reluctant to use corn-based ingredients such as CGM and CPC due to the possibility of causing yellowing of fish fillets in fish species prized for their white flesh. In addition, the impact of CPC inclusion on nutrient digestibility is currently not known and will be investigated in the present study. The objectives of this experiment were thus to investigate the effects of feeding graded levels of CPC on (a) growth performance and nutrient utilization efficiency, (b) body composition, (c) skin and fillet coloration and (d) body-organ indices and gut histology.

2 | MATERIALS AND METHODS

2.1 | Experimental fish and system

Red hybrid tilapia, *Oreochromis* sp., fingerlings were purchased from a local fish farm (Sungai Petani, Kedah, Malaysia) and held in 1,000-L fibreglass tanks upon arrival at Fish Nutrition Laboratory, Universiti Sains Malaysia. Fish were acclimated to laboratory conditions for 2 weeks and fed a commercial tilapia diet (Cargill Malaysia; 300 g/kg crude protein). Fifteen 95-L rectangular-shaped glass aquaria (30 cm width × 70 cm length × 46 cm height) were prepared for stocking the fish and mounted in a water flow-through system. Sand-filtered, dechlorinated public utility water was flowed through the aquaria at a flow rate of 15 L/hr. An air stone was placed in each aquarium for continuous aeration. The system was housed within a room illuminated with fluorescent lights, which were automatically set for continuous 14-hr light per day. Water quality parameters were measured using a portable multi-probe CyberScan PCD 650 water quality meter (Eutech Instruments); the range of water temperature, pH and dissolved oxygen were at 28.0 ± 1.0°C, 7.0 ± 0.2 and 6.0 ± 0.1 mg/L, respectively, throughout the study. At the start of the experiment, each tank was stocked with 15 fish (mean individual initial weight, 10.33 ± 0.03 g).

2.2 | Experimental diets and feeding

A total number of five isonitrogenous (350 g/kg crude protein), isolipidic (100 g/kg crude lipid) and isocaloric (14.65 MJ/kg diet) practical diets were formulated to contain graded levels of CPC (Lysto[®]) that substituted 0% (referred to as CPC0), 25% (referred to as CPC25), 50% (referred to as CPC50), 75% (referred to as CPC75) or 100% (referred to as CPC100) FM (Table 2). High levels of FM were used in the basal diet to allow us to fully evaluate the nutritive value of CPC at high dietary levels when used to replace FM. The finely ground ingredients were weighed and mixed with Hobart mixer (Liang Traco) for 30 min

TABLE 2 Ingredients and proximate composition (g/kg dry diet) of experimental diets containing graded levels of corn protein concentrate (CPC) that substituted fishmeal

Ingredients (g/kg diet)	Experimental diets ^a				
	CPC0	CPC25	CPC50	CPC75	CPC100
Fishmeal ^b	479.4	359.5	239.7	119.8	0.0
Corn protein concentrate ^c	0.0	105.8	211.7	317.5	423.4
Fish oil	1.3	14.7	28.2	41.6	55.0
Corn oil	55.0	41.3	27.7	14.0	0.3
Corn starch	277.4	274.2	271.0	267.8	264.6
Vitamin mix ^d	30.0	30.0	30.0	30.0	30.0
Mineral mix ^e	20.0	20.0	20.0	20.0	20.0
Dicalcium phosphate	10.0	10.0	10.0	10.0	10.0
Carboxymethyl cellulose	15.0	15.0	15.0	15.0	15.0
Chromic oxide	5.0	5.0	5.0	5.0	5.0
α -cellulose	106.9	124.3	141.8	159.2	176.7
Proximate composition					
Moisture	127.9	115.3	107.7	102.3	113.8
Crude protein	363.2	359.6	351.8	357.7	351.3
Crude lipid	104.7	100.9	99.1	92.6	90.6
Ash	106.0	89.0	70.5	55.5	37.1
Fibre	20.4	20.6	20.9	18.4	24.9
Nitrogen-free extract ^f	405.8	429.9	457.7	475.8	495.9

^aCPC denotes corn protein concentrate. The number following the letters denotes the per cent of fishmeal which is substituted by CPC, respectively, to each diet. ^bDanish fishmeal purchased from Liang Traco (730 g/kg crude protein). ^cLysto[®] supplied by Cargill Corn Milling (826.7 g/kg crude protein). ^dVitamin premix (g/kg of premix): ascorbic acid, 45; inositol, 5; choline chloride, 75; niacin, 4.5; riboflavin, 1; pyridoxine.HCl, 1; thiamin mononitrate, 0.92; calcium d-pantothenate, 3; retinyl acetate, 0.6; cholecalciferol, 0.083; menadione sodium bisulfide, 1.67; dl- α -tocopheryl acetate (500 IU/g), 8; d-biotin, 0.02; folic acid, 0.09; vitamin B12, 0.00135; cellulose, 845.11. ^eMineral premix (g/kg of premix): calcium phosphate.H₂O, 270.98; calcium lactate, 327; ferrous sulphate. H₂O, 25; magnesium sulphate.7H₂O, 132; potassium chloride, 50; sodium chloride, 60; potassium iodide, 0.15; copper sulphate.5H₂O, 0.785; manganese oxide, 0.8; cobalt carbonate, 1; zinc oxide, 3; sodium selenite.5H₂O, 0.011; calcium carbonate, 129.27. ^fNitrogen-free extract = 100% - (crude protein + crude lipid + ash + fibre).

for homogeneity. Chromic oxide was added as an inert marker for diet digestibility measurements. Oil was slowly added after the ingredients were thoroughly mixed. Fish oil and corn oil were added proportionally in all diets at a final ratio of 1:1 after taking into consideration the residual oils present in FM and CPC, respectively. Thereafter, distilled water was added into the mixture to achieve a moist dough. This dough was then pelleted using a locally assembled meat mincer (Liang Traco), and the resultant feed pellets of about 2-mm size were fan-dried in a drying cabinet at room temperature. The diets were sealed into labelled zipper bags and stored at -20°C until use.

Fish were hand-fed twice daily, to apparent satiation, in two equal meals at 0900 and 1600 hr, for 63 days. Each diet was fed to triplicate tanks, in a completely randomized experimental design.

2.3 | Sample collection and analysis

At the start of the feeding trials, 20 fish were randomly selected from the acclimating tank and starved for 24 hr to empty their digestive

tracts. Then, these fish were killed by an overdose of tricaine methanesulphonate (MS-222; Sigma-Aldrich) and stored at -20°C for initial whole-body composition analysis.

During the feeding trial, fish from each experimental tank were bulk-weighed every week to monitor growth performance. Four weeks before the end of the feeding trial, any uneaten feed and faecal residues were siphoned out from the tank bottom 1 hr after each feeding and discarded. New faecal materials were then carefully siphoned and collected with a fine mesh net. Only intact strands of faecal materials were collected. Tilapia faeces are expelled in long strands encased within a clear gelatinous material. As long as the faecal strands are intact, the effect of leaching on nutrient digestibility estimates was believed to be minimal (Bahurmiz & Ng, 2007). Faecal samples collected from each tank were pooled, kept in freezer at -20°C before being oven-dried and finely ground for analysis (proximate composition and Cr₂O₃ concentration).

At the end of the experiment, all fish from each tank were starved for 24 hr, euthanized by an overdose of MS-222, individually

weighed and their total body length measured. Blood samples were collected from six fish per tank into capillary tubes for subsequent determination of their haematocrit values using a micro-haematocrit centrifuge (Pico 17; Thermo Scientific). Then, tissue samples such as liver, intestine, gonad and fat were excised and weighed for the calculation of the hepatosomatic index (HSI), viscerosomatic index (VSI), gonadosomatic index (GSI) and intraperitoneal fat value (IPF). In addition, three fish were collected from each tank for total carotenoids measurement in the whole skin and pooled representative portions of the ventral muscle. Skin and flesh coloration analysis was conducted in the middle portion of the fish body. Representative portions of the posterior gut were used for histological analysis.

2.4 | Chemical, carotenoids and colour analysis

Proximate composition including moisture content, ash, crude protein and crude fibre of feed ingredients, experimental diets, whole-body samples and faeces was determined according to the standard methods of AOAC (1997) procedures as follows: moisture content was calculated from the weight loss from the sample after drying in oven at 100°C for 24 hr. Ash was determined from the weight remained after the sample was heated to 550°C for 5 hr in furnace. Nitrogen was determined by using Kjeldahl method, and protein was estimated by multiplying the nitrogen content by 6.25. Total crude lipid content was determined by chloroform/methanol extraction method described by Bligh and Dyer (1959). Concentrations of Cr_2O_3 in the diets and faeces were determined through the wet-acid digestion method described by Furukawa and Tsukahara (1966). The apparent digestibility coefficient (ADC) for dry matter, total protein and lipid of diet on dry weight basis was calculated according to NRC (2011).

Skin and muscle colorations were determined using a portable Minolta colour reader instrument (CR-100). Target or reference sample was determined by the calibration plate (white colour display), which was provided along with the instrument. *L*-value indicates lightness of the sample, with darkest black at *L* = 0 and brightest white at *L* = 100. Redness coloration was determined by the *a*-value, while yellowish coloration was determined by the *b*-value.

Total carotenoid in muscle samples was measured with the method described by Barua, Kostic, and Olson (1993) and calculated according to Bell et al. (2001). Briefly, each muscle sample was collected and chopped into small pieces. A total amount of 1 g of sample was homogenized in 5 ml of absolute ethanol and 5 ml of ethyl acetate for 2 min. The homogenate was then centrifuged at the speed of 5,000 rpm for 5 min, and the supernatant was transferred to a stopped glass tube. The pellet was re-homogenized in 5 ml of ethyl acetate and re-centrifuged. The supernatant was combined with the first supernatant. Next, the pellet was again re-homogenized in 10 ml of hexane and re-centrifuged. The supernatant was then combined with the previous supernatants. The pooled supernatant was then dried under a stream of nitrogen and vacuum-desiccated for 2 hr, and the residue was re-dissolved in 2 ml of hexane containing 0.2 g/L butylated hydroxyl toluene. Finally, the residue was centrifuged at the speed of 5,000 rpm for 5 min and the total carotenoid

of the samples measured spectrophotometrically at 470 nm using E1% (wt/v) of 2,100 (Bell et al., 2001).

2.5 | Gut histology

Gut samples were collected during final sampling and fixed in Bouin's solution for 24 hr. The samples were then processed, embedded in paraffin wax, sectioned using a microtome knife at the thickness of 8 μm and stained with haematoxylin and eosin using routine histological procedures. The slides were examined under compound microscope (Model: BX41-CCD; Olympus) at two different magnifications, 4 $\times$ and 10 $\times$. The structural changes in the lamina propria, submucosa and tunica muscularis of fish from the different treatments were examined qualitatively.

2.6 | Statistics analysis

Results were expressed as mean $\pm$ standard error ($n = 3$). Each aquarium was considered an experimental unit. Data were analysed with one-way analysis of variance (ANOVA) to determine the possible significant differences among the dietary treatments, at the significance level of 0.05. Differences among means were further analysed using Duncan's multiple range test. Polynomial regression was used to determine the optimal CPC level for growth and feed utilization parameters (Yossa & Verdegem, 2015). The statistical analysis was performed using SPSS statistics 17.0, for Windows (SPSS Inc.).

3 | RESULTS

3.1 | Growth and feed utilization

There was a significant effect of the treatments on final body weight (FBW), WG, SGR, FI, FCR, PER and survival (Table 3: $p < 0.05$). The FBW and WG increased when CPC replaced 25% of FM in the diet (CPC25), but a further inclusion of CPC in the diet resulted in gradual decrease in FBW and WG. Polynomial analysis showed quadratic regressions with FBW ($y = -0.0082x^2 + 0.4245x + 57.445$; $R^2 = 0.98$) and WG ($y = -0.0792x^2 + 4.0844x + 456.6$ $R^2 = 0.9798$); using these regressions, the optimum substitution levels of FM by CPC were estimated at 25.88% and 25.78%, for FBW and WG, respectively. There was no significant change ($p > 0.05$) in the SGR with the increase in the substitution of FM with CPC up to 50% (CPC50); from this level, a further increase in the dietary CPC level led to a gradual decrease in SGR until CPC100. Polynomial analysis showed a quadratic regression with SGR ($y = -0.0004x^2 + 0.0234x + 2.6592$ $R^2 = 0.9459$); using this regression, the optimum substitution levels of FM by CPC were estimated at 29.25%. Fish survival was generally not significantly different between treatments with the highest survival observed in fish fed the CPC25 diet. There was no change in the FI and FCR with the increase in the substitution of FM with CPC up to 75% (CPC75); from this level, the further increase in the CPC level (CPC100) led to significant decrease in FI and increase in FCR.

TABLE 3 Growth performances and feed utilization efficiency of red hybrid tilapia, *Oreochromis* sp., fingerlings fed diets containing graded levels of corn protein concentrate (CPC) for 63 days

	Diets ^b				
	CPC0 (control)	CPC25	CPC50	CPC75	CPC100
IBW ^c (g)	10.32 ± 0.02	10.33 ± 0.00	10.33 ± 0.03	10.33 ± 0.02	10.31 ± 0.02
FBW ^d (g)	57.84 ± 1.87 ^c	62.72 ± 0.95 ^d	56.37 ± 0.62 ^c	45.68 ± 1.12 ^b	16.82 ± 1.49 ^a
WG ^e (%)	460.4 ± 17.4 ^c	507.2 ± 9.4 ^d	445.4 ± 6.4 ^c	342.2 ± 11.9 ^b	63.1 ± 14.4 ^a
SGR ^f (%BW/day)	2.73 ± 0.05 ^c	2.86 ± 0.02 ^c	2.69 ± 0.02 ^c	2.36 ± 0.04 ^b	0.76 ± 0.15 ^a
Survival ^g (%)	91.1 ± 4.4 ^{ab}	97.8 ± 2.2 ^b	84.4 ± 2.2 ^{ab}	75.5 ± 5.8 ^a	88.8 ± 8.0 ^{ab}
FI ^h (g/fish)	3.40 ± 0.13 ^b	3.44 ± 0.04 ^b	3.32 ± 0.02 ^b	3.20 ± 0.06 ^b	2.50 ± 0.08 ^a
FCR ⁱ	1.11 ± 0.01 ^a	1.07 ± 0.01 ^a	1.12 ± 0.01 ^a	1.31 ± 0.05 ^a	3.59 ± 0.24 ^b
PER ^j	2.51 ± 0.04 ^c	2.64 ± 0.05 ^c	2.59 ± 0.03 ^c	2.28 ± 0.06 ^b	0.86 ± 0.17 ^a

^aValues are the mean ± SE of triplicate groups of fish. Mean values within same row with the different superscripts are significantly different ($p < 0.05$). ^bSee Table 1 footnote for diet description. ^cInitial body weight (g). ^dFinal body weight (g). ^eWeight gain (%) = [(FBW – IBW)/IBW] × 100. ^fSpecific growth rate (%BW/day) = {ln (FBW) – ln (IBW)} / feeding days × 100. ^gSurvival (%) = (Final fish number – Initial fish number) × 100. ^hFeed intake (g/fish) = Total dry feed intake/Number of fish over 63 days. ⁱFeed conversion ratio = Total dry feed intake (g)/Wet weight gain (g). ^jProtein efficiency ratio = Weight gain (g)/Total protein intake (g).

Polynomial analysis showed quadratic regressions with FI ($y = -0.0002x^2 + 0.0088x + 3.367$; $R^2 = 0.8744$) and FCR ($y = 0.0005x^2 - 0.0338x + 1.283$; $R^2 = 0.8824$); using these regressions, the optimum substitution levels of FM by CPC were estimated at 22.00% and 33.80%, for FI and FCR, respectively. There was no change in the PER with the increase in the substitution of FM with CPC up to 50% (CPC50); from this level, a further increase in the CPC level led to a gradual decrease in PER until CPC100. Polynomial analysis showed quadratic regression with SGR ($y = -0.0004x^2 + 0.0235x + 2.4317$; $R^2 = 0.9257$); using this regression, the optimum substitution levels of FM by CPC were estimated at 29.37%.

3.2 | Haematocrit and body-organ indices

The effect of the treatments was significant on the haematocrit value, CF, and HSI, and not significant on IPF, VSI, and GSI of both

the male and the female fish (Table 4). Fish fed the control diet clearly showed a higher blood haematocrit value as compared with those fed the CPC75 diet. The CF remained unchanged with the substitution of FM with CPC until the level of 50%, after which the CF gradually decreased with the increase in CPC in the diets. The HSI increased with the substitution of FM with CPC until the level of 75%, after which the HSI decreased (CPC100).

3.3 | Whole-body composition

The effect of the treatments was significant on whole-body moisture, crude protein and ash and not significant on whole-body crude lipid content of fish (Table 5). Fish fed the control diet clearly showed a higher whole-body dry matter and crude protein content as compared with those fed the CPC100 diet. There was no significant difference on whole-body ash content among the fish fed the control

TABLE 4 Haematocrit, condition factor and body-organ indices of red hybrid tilapia, *Oreochromis* sp., fingerlings fed diets containing graded levels of corn protein concentrate (CPC) for 63 days

	Diets ^b				
	CPC0	CPC25	CPC50	CPC75	CPC100
Haematocrit ^c (%)	39.00 ± 1.57 ^b	37.00 ± 0.15 ^{ab}	37.00 ± 0.86 ^{ab}	34.00 ± 0.24 ^a	37.00 ± 1.56 ^{ab}
CF ^d (g/cm ³)	1.70 ± 0.01 ^c	1.72 ± 0.02 ^c	1.71 ± 0.03 ^c	1.65 ± 0.02 ^b	1.51 ± 0.02 ^a
HSI ^e	2.23 ± 0.16 ^{ab}	2.01 ± 0.05 ^{ab}	2.41 ± 0.19 ^b	2.45 ± 0.19 ^b	1.76 ± 0.10 ^a
IPF ^f	0.32 ± 0.04	0.33 ± 0.01	0.29 ± 0.03	0.41 ± 0.05	0.46 ± 0.11
VSI ^g	2.59 ± 0.08	2.93 ± 0.21	2.66 ± 0.16	3.05 ± 0.15	3.04 ± 0.14
GSI (♂) ^h	1.09 ± 0.04	1.04 ± 0.08	1.09 ± 0.08	1.33 ± 0.29	1.53 ± 0.34
GSI (♀) ⁱ	3.69 ± 0.56	3.60 ± 0.65	3.26 ± 1.07	5.16 ± 1.01	5.02 ± 0.13

^aValues are the mean ± SE of triplicate groups of fish. Mean values within same row with the different superscripts are significantly different ($p < 0.05$). ^bSee Table 1 footnote for diet description. ^cHaematocrit (%) = (Length of haemoglobin/Total length of plasma and haemoglobin) × 100. ^dCondition factor (g/cm³) = Weight of fish/(length of fish)³ × 100. ^eHepatosomatic index = [Liver weight (g)/FBW (g)] × 100. ^fIntraperitoneal fat = [Body fat weight (g)/FBW (g)] × 100. ^gViscerosomatic index = [Viscera weight (g)/FBW (g)] × 100. ^hGonadosomatic index (♂) = [Testis weight (g)/FBW (g)] × 100. ⁱGonadosomatic index (♀) = [Ovary weight (g)/FBW (g)] × 100.

Diets ^b	Moisture	Protein	Lipid	Ash
Initial	766.8 ± 2.1	129.9 ± 1.1	53.5 ± 0.7	38.4 ± 0.2
CPC0	780.5 ± 19.3 ^b	124.7 ± 9.1 ^b	56.5 ± 7.6	34.5 ± 2.1 ^c
CPC25	787.6 ± 5.9 ^{ab}	119.8 ± 2.3 ^{ab}	53.7 ± 3.1	33.9 ± 0.9 ^b
CPC50	801.3 ± 4.4 ^{ab}	114.8 ± 2.3 ^{ab}	51.4 ± 1.1	29.5 ± 1.1 ^a
CPC75	841.0 ± 35.2 ^{ab}	111.1 ± 1.5 ^{ab}	55.9 ± 2.1	26.0 ± 0.5 ^a
CPC100	817.8 ± 10.7 ^a	106.9 ± 4.5 ^a	45.1 ± 4.1	28.2 ± 1.4 ^a

^aValues are the mean ± SE of triplicate groups of fish. Mean values within same column with the different superscripts are significantly different ($p < 0.05$).

^bSee Table 1 footnote for diet description.

or CPC25 diets, and fish from both groups had a significantly higher whole-body ash content than those fed the other treatments.

3.4 | Apparent digestibility coefficients

The inclusion of CPC was significant on diet protein and lipid digestibility (Table 6). Even though dry matter digestibility was not significantly impacted, the CPC100 diet showed the lowest dry matter digestibility at 56.9%. The control, CPC25, CPC50 and CP75 diets had similar protein digestibility with the CPC50 and CP75 diets showing a significantly higher protein digestibility than the CPC100 diet. Lipid digestibility decreased gradually with the increase in the CPC level in the diets, from 95% with CPC0 to 77% with CPC100.

3.5 | Carotenoids content and skin and muscle coloration

There was a significant difference among the treatments in terms of total carotenoid content, with fish fed CPC100 diets showing a significantly higher total carotenoid content than those fed the control diet (Table 7). There were no significant differences in coloration among the treatments in terms of lightness (L -value), E -values and the redness coloration (positive a -values) in both skin and muscle. The chroma in the fish skin and positive b -value (yellowness) in both the fish skin and muscle showed a significant increasing trend from control to CPC100 diets, while the chroma in the muscles was not significantly different among the treatments. This trend in increasing b -values of tissues when fed increasing dietary inclusion of CPC was visually indicated by the yellow colour observed in the skin and ovaries of fish fed the CPC100 diet,

compared with those fed the control diet (Figure 1). Despite significantly higher b -values in the muscle of fish fed the CPC100 diet compared to control as detected instrumentally by the colour reader (Table 7), the yellow discoloration of the fillets was not visually obvious.

3.6 | Histology

The histological analysis did not reveal any differences in the lamina propria of mucosa and submucosa among fish fed the various experimental diets (Figure 2). No significant thickening of gut wall was observed. Some uneven mucosal folded structures were observed in fish fed the CPC75 or CPC100 diets but the occurrences were not consistent and further histological examinations are warranted.

4 | DISCUSSION

The present study aimed at evaluating whether CPC alone could be considered as an alternative plant protein ingredient to replace FM in red hybrid tilapia diets. The CPC (Lysto[®]) used in this study had an improved nutritional value with higher protein and lysine levels. According to the manufacturers (Cargill Corn Milling; Cargill Inc.), lysine levels of Lysto[®] are about 52 g/kg CPC. The results of the present study showed that replacing up to 50% FM in red hybrid tilapia diet with CPC did not show any adverse effects on the growth performance, nutrient utilization efficiency, body-organ indices and gut histology of fish. Similarly, Khalifa et al. (2018) reported that up to half of the FM could be replaced by CPC in Nile tilapia diet without

TABLE 6 Apparent digestibility coefficient (% dry matter basis) of red hybrid tilapia, *Oreochromis* sp., fingerlings fed diets containing graded levels of corn protein concentrate (CPC)

	Diets ^b				
	CPC0	CPC25	CPC50	CPC75	CPC100
Dry matter	62.26 ± 1.46	63.68 ± 2.63	63.75 ± 1.96	63.65 ± 0.45	56.93 ± 4.55
Protein	80.63 ± 0.43 ^{ab}	80.92 ± 2.37 ^{ab}	83.92 ± 0.87 ^b	84.65 ± 0.97 ^b	73.86 ± 5.14 ^a
Lipid	94.91 ± 0.05 ^d	94.05 ± 0.36 ^d	91.55 ± 0.32 ^c	87.58 ± 0.35 ^b	77.64 ± 1.54 ^a

^aValues are the mean ± SE of triplicate groups of fish. Mean values within same roll with the different superscripts are significantly different ($p < 0.05$). ^bSee Table 1 footnote for diet description.

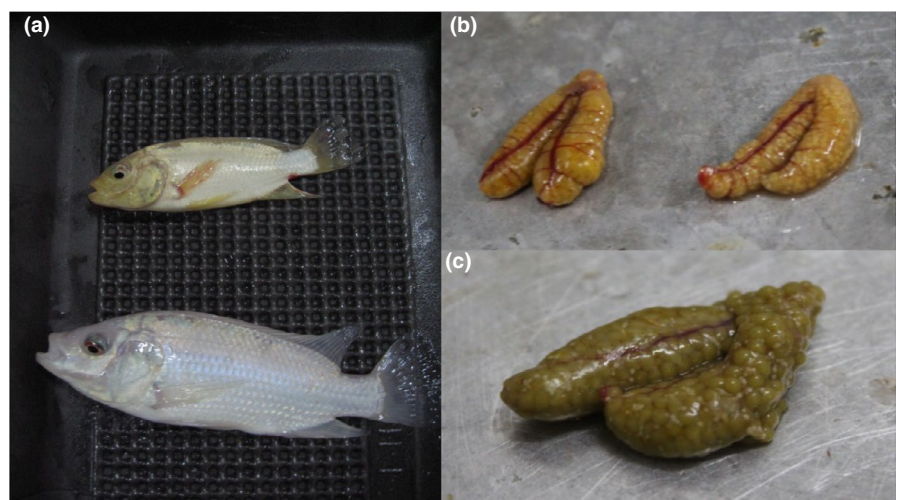
TABLE 7 Skin, muscle coloration and total carotenoids of red hybrid tilapia, *Oreochromis* sp., fingerlings fed diets containing graded levels of corn protein concentrate (CPC) for 63 days

	Diets ^b				
	CPC0	CPC25	CPC50	CPC75	CPC100
Coloration					
	<i>L</i> ^c				
Skin	24.11 ± 1.01	24.44 ± 2.05	23.12 ± 1.45	23.67 ± 0.35	24.86 ± 1.39
Muscle	46.27 ± 1.82	49.64 ± 2.09	44.25 ± 2.38	43.86 ± 2.36	44.60 ± 2.39
	<i>Chroma</i> ^d				
Skin	3.46 ± 0.38 ^a	3.48 ± 0.44 ^a	4.29 ± 0.65 ^{ab}	5.60 ± 0.42 ^b	7.48 ± 0.07 ^c
Muscle	3.78 ± 0.37	4.13 ± 1.25	4.29 ± 1.28	4.49 ± 0.64	4.22 ± 0.72
	<i>h</i> ^{oe}				
Skin	1.59 ± 0.43 ^a	2.04 ± 0.42 ^a	2.24 ± 0.54 ^{ab}	2.99 ± 0.18 ^{bc}	3.42 ± 0.04 ^c
Muscle	1.12 ± 0.23 ^a	1.02 ± 0.51 ^a	1.39 ± 0.70 ^{ab}	1.77 ± 0.43 ^{ab}	2.02 ± 0.22 ^b
	<i>E</i> ^f				
Skin	24.82 ± 1.19	24.92 ± 2.10	23.78 ± 1.27	24.67 ± 0.43	26.33 ± 1.31
Muscle	46.49 ± 1.85	49.91 ± 2.15	44.91 ± 2.31	44.22 ± 2.36	44.92 ± 2.32
	<i>a</i> ^g				
Skin	1.11 ± 0.13	0.69 ± 0.79	1.94 ± 0.34	1.34 ± 0.24	1.11 ± 0.51
Muscle	2.84 ± 0.61	3.27 ± 1.52	2.59 ± 1.41	3.07 ± 0.88	2.36 ± 0.68
	<i>b</i> ^h				
Skin	2.54 ± 0.93 ^a	3.76 ± 0.31 ^a	3.87 ± 1.20 ^a	6.05 ± 0.51 ^b	7.89 ± 0.06 ^b
Muscle	1.62 ± 0.09 ^{ab}	1.28 ± 0.96 ^a	2.56 ± 0.87 ^{abc}	2.84 ± 0.75 ^{bc}	3.66 ± 0.58 ^c
Total carotenoids (mg/kg)					
Muscle	0.016 ± 0.01 ^a	0.028 ± 0.00 ^{ab}	0.026 ± 0.00 ^{ab}	0.026 ± 0.00 ^{ab}	0.035 ± 0.01 ^b

^aValues are the mean ± SE of triplicate groups of fish. Mean values within same roll with the different superscripts are significantly different ($p < 0.05$). ^bSee Table 1 footnote for diet description. ^c ΔL denotes lightness and shows degree of lightness. ^d Δ chroma shows the vividness or dullness of colour. ^e Δh° denotes Hue, units described in the form of degree ($^\circ$), ranging from 0° (red) through 90° (yellow), 180° (green), 270° (blue) and black to 0° . ^f ΔE shows differences between samples colour and reference sample. ^g Δa represents coloration more to green when a -value shows negative and more to red when a -value shows positive. ^h Δb represents coloration more to blue when b -value shows negative and more to yellow when b -value shows positive.

any adverse effects on fish growth. However, it should be noted that in the study of Khalifa et al. (2018), for the basal diet, FM only provided 144 g/kg protein with the remaining dietary protein (206 g/

kg) coming from SBM. Therefore, even at 100% replacement of FM, CPC in the study of Khalifa et al. (2018) contributed at the most, about 15% of the total protein of the experimental diets. Despite

FIGURE 1 The visual coloration of red hybrid tilapia, *Oreochromis* sp., fingerlings fed diets containing graded levels of corn protein concentrate (CPC) for 63 days. Fish fed the CPC100 diet were smaller at the end of the feeding trial and showed yellowish skin (a-top) and ovary colours (b) while those fed the control diet (a-bottom) showed usual pinkish skin (a-bottom) and greenish ovary (c) colours

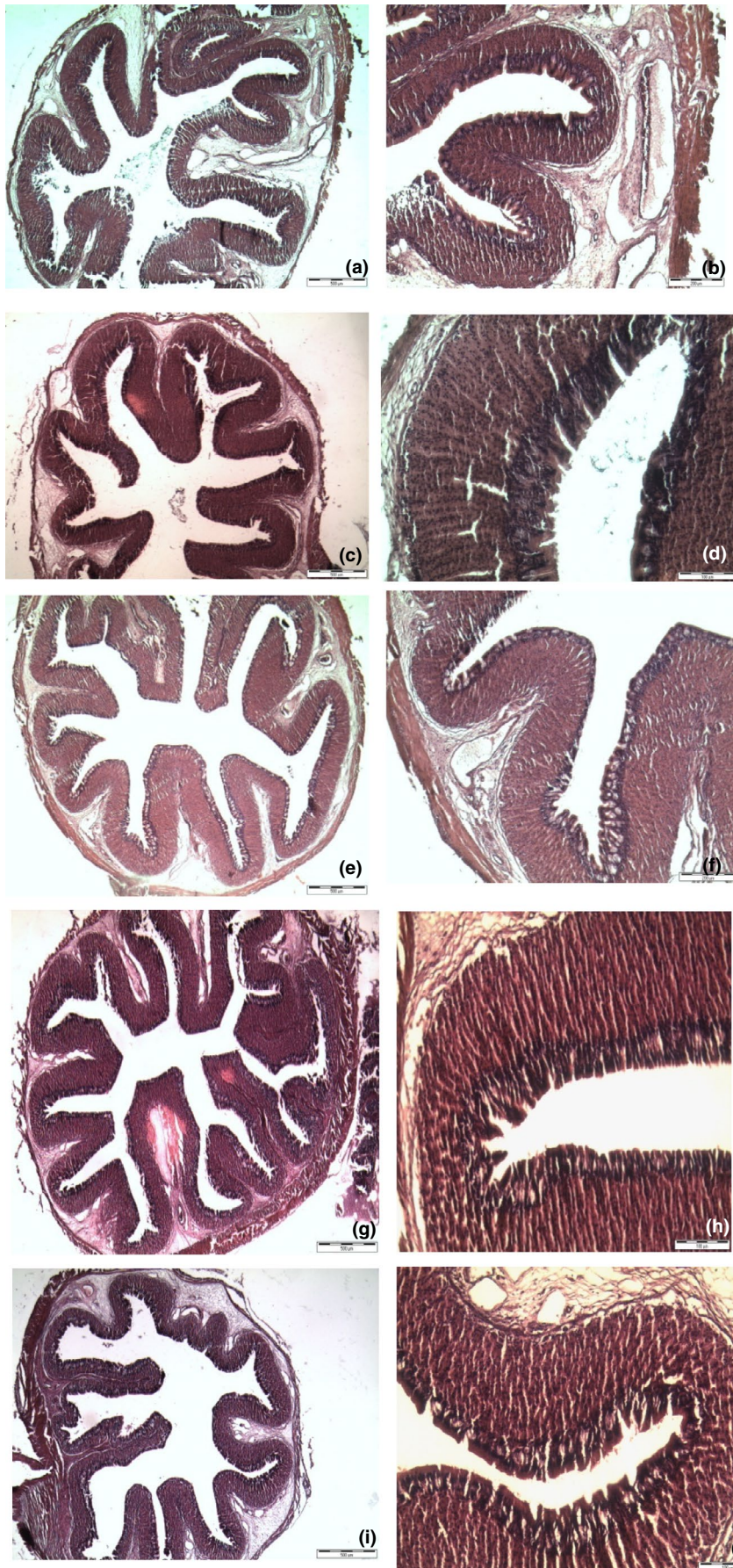


FIGURE 2 Histology of gut of red hybrid tilapia, *Oreochromis* sp., fed diets containing graded levels of corn protein concentrate (CPC) for 63 days. (a, b) The histology of the fish fed the control (CPC0) diet at the 4× and 10× magnification, respectively; (c, d) for the CPC25 diet at the 4× and 10× magnification, respectively; (e, f) for the CPC50 diet at the 4× and 10× magnification, respectively; (g, h) for the CPC75 diet at the 4× and 10× magnification, respectively; and (i, j) for the CPC100 diet at the 4× and 10× magnification, respectively. The microscope scale of the histology slides was 5.5 cm: 500 μm in ratio at 4× magnification and 5.5 cm: 200 μm in ratio for 10× magnification

having similar dietary protein levels, the present study contained only two protein sources, FM and CPC, and therefore, it was possible to investigate the impact of higher dietary protein contributions by CPC in tilapia diets and better elucidate the nutritive value of CPC without the compounding effects of other plant protein sources. At the highest level, CPC was incorporated at 423.4 g/kg diet in the present study compared to only 190 g/kg diet in the study of Khalifa et al. (2018). The present study showed that CPC could be included at up to 211.7 g/kg diet (CPC50) without any negative impact on fish growth compared to only 100 g/kg CPC as reported by Khalifa et al. (2018). Apart from potential fish species and diet formulation differences, the difference in fish size used might have also contributed to the larger capacity of red hybrid tilapia to utilize CPC as a protein source. Furthermore, to estimate the optimum level of FM replacement with CPC for tilapia, regression analyses were performed on weight gain, FCR and PER which showed that the optimal substitution level of FM by CPC is expected to be between 25% and 33%.

Direct comparisons of the results of the present study with other published studies on the use of CPC in fish feeds, with the exception of Khalifa et al. (2018), are not possible since CPC was used as a part of a protein blend to replace FM in the diets of Atlantic salmon (Burr et al., 2012) and Florida Pompano (Cook et al., 2016; Novriadi et al., 2019). Nevertheless, similar successful partial replacement of FM with corn protein sources has been reported in other studies; for instance, CGM successfully substituted FM in the diets of turbot, *Psetta maxima* (Regost, Arzel, & Kaushik, 1999), two-banded seabream, *Diplodus vulgaris* (Bulut et al., 2014) and Nile tilapia (El-Ebiary, 2005). The weight gain observed in the present study with red hybrid tilapia fed the CPC25 diet ($507.26 \pm 9.39\%$) was higher than 468.28% or 405.31% reported by El-Ebiary (2005) with Nile tilapia fed diet in which 25% FM was substituted with CGM or SBM, respectively. Several studies have reported that the imbalance of amino acids profile in CGM, especially methionine and lysine, restricted the feed intake and consequently reduced growth of fish fed diets with higher inclusion levels of CGM (D'Mello, 1993; Goda et al., 2007). With regard to other plant protein concentrates, generally, greater inclusion levels can be used in fish feeds compared to their less processed forms (Oujifard et al., 2012; Slawski et al., 2012; Tusche et al., 2013; Zhang et al., 2012; Zhao et al., 2010). Total replacement of FM with rapeseed protein concentrate was reported possible for rainbow trout (Slawski et al., 2012) and with soy protein concentrate for Nile tilapia (Zhao et al., 2010) as long as the essential amino acid requirements of the target fish species were met.

The results of the present study showed that replacing 75% or 100% FM with CPC has deleterious effects on growth, feed utilization efficiency, haematocrit and body-organ indices of red hybrid tilapia, especially at the 100% replacement level. It is generally known that higher substitution levels of FM by plant ingredients could result in depressed growth performance of aquatic animals (Gatlin et al., 2007). Since the CPC used in the present study is not known to contain ANF, it is likely that the depressed fish growth observed in fish fed CPC75 or CPC100 diet was due to lowered nutrient digestibility and/or feed intake. The CPC100 diet showed the lowest ADC for

dry matter, protein and lipid among all the experimental diets. The significantly lowered availability of nutrient intake was reflected in the whole-body of fish which showed lower levels of body protein and lipid when fed the CPC100 diet. The gradual decrease in the whole-body protein content of tilapia fed with increasing inclusion of CPC in the diet suggested that CPC might not provide as much metabolizable protein as FM. Wu et al. (1995) mentioned that corn has an undesirable flavour that possibly limits its high inclusion in feeds. In the present study, the reduced feed intake observed which was significant at the 100% replacement of FM with CPC was probably due to reduced feed palatability. Further studies are currently being planned to utilize feed attractants to explore whether feed palatability issues can be mitigated in high CPC-based diets. Even though the Lysto® CPC used in the present study was enhanced with lysine, it is currently not known whether the inclusion of high dietary concentrations of CPC in fish feeds may cause deficiencies in other essential amino acids. Further studies are currently being planned to blend CPC with other plant protein sources that can complement their amino acid profiles. In contrast to Khalifa et al. (2018), preliminary observations in the present study did not reveal any major alterations in gut morphology when tilapia was fed high CPC levels in their diets.

Skin and muscle colours of fish are mainly regulated by dietary carotenoids (Chatzifotis et al., 2005). Red hybrid tilapia is a much sought-after fish, especially in Asia, due to its similar appearance (coloration) to high-value marine fish such as red snappers. As such, it is sold at premium prices compared to normal black-coloured tilapia such as Nile tilapia. In the present study, carotenoids in CPC contributed to skin yellowness, which increased with increasing CPC level in the diet (Table 7; Figure 1). The yellowness of the skin of red hybrid tilapia could reduce its market acceptance and value when sold as live or whole fish. The gradual accumulation of carotenoids in the muscle of red hybrid tilapia when fed increasing CPC in the diet also resulted in increased yellowness of fish fillets (as detected instrumentally), but this was not as visually apparent compared to changes in skin coloration. It is likely that at lower inclusion levels of CPC, visual changes in fillet colour might not be obvious and the tilapia fillet can still be marketed as white flesh fish. However, longer term feeding trials would be needed to confirm this. Khalifa et al. (2018) reported a significant change in tilapia fillet colour when fed the highest CPC (190 g/kg) diet. Other corn-based aquafeed ingredients such as yellow corn and CGM are known to induce skin and/or fillet yellowness in fish (Hu, Ferrell, Lim, & Davis, 2012; Park, Flores, & Johnson, 1997). Hu et al. (2012) found that pond-raised hybrid catfish fed with 20% CGM had significantly higher levels of yellow pigments in fillet, with regard to lutein, zeaxanthin and total xanthophylls content, than fish fed the control diet.

In conclusion, the results of the present study showed that replacing up to 50% FM with CPC in red hybrid tilapia diet did not show any adverse effects on the growth performance, nutrient utilization efficiency, body-organ indices, whole-body composition, gut histology and tissue coloration of red hybrid tilapia. Regression analysis estimated the optimal substitution level of FM by CPC at 25% for weight gain, 33% for FCR and 29% for protein efficiency. Plant



protein sources such as CPC when used to replace marine-based protein sources such as FM can contribute to the continued sustainability and scalability of modern-day tilapia farming. Further studies are needed on using CPC in cost-effective and sustainable tilapia diets.

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